

Electric circuits virtual laboratories teacher guide

Year 10 GCSE Physics

This guide follows the student booklet. Each activity keeps simulator actions, observations, measurements and calculations in separate sections to reduce working-memory demand.

TEACHING SEQUENCE

Lesson section	Suggested time	Main learning
Setup and meters	10 minutes	Build the loop and place probes
Lab 1	35 minutes	Current, charge and $I = Q \div t$
Lab 2	45 minutes	Potential difference, energy and $V = E \div Q$
Lab 3	40 minutes	Resistance and $V = IR$
Core check	10 minutes	Independent retrieval
Challenge	15 to 25 minutes	Reasoning and circuit design

Print the setup and meter pages once per pupil. The three labs can be issued separately. Keep the challenge pages until the core work is secure.

PREPARATION

Provide one device per student, calculators, pencils and rulers. Demonstrate how to join wire ends, edit component values, use the Non Contact Ammeter and place voltmeter probes. Use minimum wire resistivity and battery resistance throughout.

[Open PhET Circuit Construction Kit DC](#)

LANGUAGE TO REINFORCE

Say “charge passes a point”, “energy is transferred” and “potential difference is measured across two points”. Avoid saying that current is used up, voltage flows or a battery creates electrons.

Lab 1 answers and teaching notes

Current and charge

CURRENT AROUND THE LOOP

With a 6 V battery and 10 Ω resistor, the ideal current is approximately 0.60 A at A, B and C.

Breaking the loop gives zero steady current. Reconnecting the loop restores the reading.

Electrons move from the negative terminal towards the positive terminal in the external circuit.

Conventional current points from positive to negative. Changing the display does not change the current magnitude.

EXPECTED CONCLUSION

Charge continues around the complete loop and is conserved. The resistor transfers energy from the electrical pathway to its thermal energy store and then to the surroundings. It does not use up charge or current.

CURRENT READINGS

Battery setting / V	Expected current I / A
3	0.30
6	0.60
9	0.90

CHARGE CALCULATIONS

Battery / V	Current I / A	Q in 10 s / C	Q in 20 s / C
3	0.30	3.0	6.0
6	0.60	6.0	12
9	0.90	9.0	18

At constant current, doubling time doubles the charge passing a point. Waiting longer does not itself change the current.

QUESTIONS WHILE CIRCULATING

Ask: "Where is the point at which charge is passing?" "Which value came from the meter?" "Which value did you calculate?" "If current were used up, why are the readings equal?"

Lab 2 answers and teaching notes

Potential difference and energy

ONE RESISTOR RESULTS

Battery setting / V	Battery p.d. / V	Resistor p.d. / V
3	about 3	about 3
6	about 6	about 6
9	about 9	about 9

With both probes on the same ideal wire, the reading should be approximately 0 V. Across the resistor at a 6 V setting, it should be approximately 6 V. Potential difference compares two points.

ENERGY CALCULATIONS

Resistor p.d. V / V	Charge Q / C	Expected energy E / J
3	2	6
6	2	12
9	2	18

A p.d. of 9 V means 9 J is transferred for each coulomb passing through the component.

TWO RESISTOR RESULTS

For two equal $10\ \Omega$ resistors in series with a 6 V battery: the current is approximately 0.30 A throughout, and each resistor has approximately 3 V across it.

Place	p.d. / V	Q / C	Energy / J
Battery supplies	6	2	12
R1 transfers	3	2	6
R2 transfers	3	2	6

The battery supplies about 12 J and the two resistors transfer about 12 J in total. The same 2 C passes through both series resistors; the charge is not divided between them.

Lab 2 comparison and misconceptions

Distinguishing current from potential difference

EXPECTED COMPARISON

Circuit	Current I / A	Resistor p.d. V / V	Energy per C
A 6 V and 10 Ω	about 0.60	about 6	about 6 J per C
B 12 V and 20 Ω	about 0.60	about 12	about 12 J per C

The circuits have approximately the same current, so the same charge passes a point each second. Circuit B transfers about twice as much energy per coulomb. Equal current does not imply equal p.d.

COMMON MISCONCEPTIONS

Student statement	Prompt to use
Voltage flows through the circuit	What does the voltmeter compare between two points?
The battery makes the charges	Where were the charge carriers before the switch was closed?
Each resistor receives half the charge	Compare current before and after each resistor.
Zero current means zero battery p.d.	Can a battery have p.d. when the loop is open?

Lab 3 answers and teaching notes

Resistance and the V I relationship

FIXED 10 Ω RESISTOR

Resistor p.d. V / V	Expected current I / A	$V \div I$ / Ω
2	0.20	10
4	0.40	10
6	0.60	10
8	0.80	10
10	1.00	10
12	1.20	10

The graph should be a straight line close to the origin. This supports direct proportionality between V and I when resistance and physical conditions remain constant. With V vertically and I horizontally, the gradient is resistance, about 10 Ω .

CHANGING RESISTANCE

Resistance R / Ω	Expected p.d. V / V	Expected current I / A
5	about 6	1.20
10	about 6	0.60
20	about 6	0.30

At fixed p.d., increasing resistance decreases current. Doubling resistance halves current for these ideal resistor settings.

CORE CHECK ANSWERS

1. A current of 0.40 A means 0.40 C passes a point each second. 2. A p.d. of 8 V means 8 J is transferred per coulomb. 3. $V = IR = 0.20 \times 15 = 3.0$ V. 4. Energy is transferred to the surroundings.

Challenge answers

Reasoning and circuit design

REASONING

1. Current readings are approximately equal before and after the resistor. Charge continues around the loop; energy is transferred in the resistor.
2. Equal current means the same charge passes each second. The component with 12 V across it transfers 12 J per coulomb, twice the 6 J per coulomb transferred by the component with 6 V across it.
3. A straight line through or close to the origin shows that doubling current doubles p.d. The ratio $V \div I$ remains approximately constant.

CIRCUIT DESIGN

Required current: $I = Q \div t = 3.0 \div 5.0 = 0.60 \text{ A}$.

Required p.d.: $V = E \div Q = 18 \div 3.0 = 6.0 \text{ V}$.

Required resistance: $R = V \div I = 6.0 \div 0.60 = 10 \Omega$.

Students should build a 6 V battery with a 10 Ω resistor. Their measured current should be about 0.60 A. Their checks should give $Q \approx 3.0 \text{ C}$ in 5.0 s and $E \approx 18 \text{ J}$.

ASSESSMENT GUIDANCE

Accept small differences caused by rounding or non-zero model settings. Require units and a clear distinction between meter readings and calculated values. Ask pupils who finish early to predict a different combination of V and R that gives the same current.

Support and troubleshooting

Practical responses during the virtual laboratories

SUPPORT FOR LEARNING NEEDS

Issue one lab at a time. Read the blue DO steps aloud before pupils begin. Ask pupils to tick each completed action. Cover later calculation columns until the measurements are finished. Provide the first reading or calculation only when a pupil cannot start independently.

Use the same spoken routine every time: "What do you change? What do you keep the same? Which meter do you read? Where do you write it?"

TROUBLESHOOTING

Problem	Check
Current reads zero	Look for an open connection or a probe that is not on a wire.
Voltmeter is negative	Swap the two probes.
Readings do not match expectations	Check battery, resistor, extra branches and Advanced Options.
Table contains calculated values too early	Return to the green page and record only meter readings.
Pupil counts moving dots	Remind them that the dots are not a calibrated charge counter.

MODEL LIMITS

The animated charges are qualitative. The tasks measure current and p.d.; charge and energy are calculated from stated intervals and quantities. The activity supports conceptual learning and does not replace practical experience with real meters or required practical work.

The sequence is aligned with the supplied Electric Circuits lesson deck. The original slide deck has not been changed.